We are delighted to welcome you all to the

Free Summer Course on AI & Ethical Hacking

organized by

SRI GANESH COLLEGE OF ENGINEERING & TECHNOLOGY

and

PSV COLLEGE OF ARTS & SCIENCE

DAY 2

PYTHON DEVELOPMENT

AGENDA

- ✓ Programming Languages
- ✓ What is Python?
- ✓ Application
- ✓ Installation

PROGRAMMING LANGUAGES

What is Programming Language?

- Programming languages are formal languages used to communicate instructions to computers.

- They are used to develop software, websites, mobile applications, and various other types of computer programs.

- Each programming language has its own syntax (rules for writing code) and semantics (meaning of the code).

TYPES OF PROGRAMMING LANGUAGE

I. Low-level languages:

These languages are closer to machine code and hardware.

Examples: *Assembly language and Machine code.*

II. High-level languages:

These languages are closer to human language and are easier for programmers to understand and use.

Examples: *Python, Java, C++, JavaScript, and Ruby.*

PYTHON

INTRODUCTION

- Python is a high-level, interpreted programming language.
- Known for its simplicity and readable syntax.
- Created by Guido van Rossum in the late 1980s.
- Used in web development, automation, data science, machine learning, etc.

Interpreted languages: **Code written in these languages is executed line by line by an interpreter.**

FEATURES OF PYTHON

- ***Easy to learn and write*** – great for beginners.
- ***Open-source*** – free to use and supported by a large community.
- ***Portable*** – runs on different platforms (Windows, Mac, Linux).
- ***Rich standard library*** – tools for file handling, math, web, etc.
- Object-Oriented & Functional Programming supported.

PYTHON USE CASES

- Web development (Django, Flask)
- Data Science & AI (Pandas, NumPy, Scikit-learn)
- Automation/Scripting (Automate tasks easily)
- Game Development (Pygame)
- Desktop Applications (Tkinter, PyQt)

INSTALLATION

- Download Python: https://www.python.org
- Run the Installer.
- Customize Installation (Optional).
- Complete Installation.
- Verify Installation.

SAMPLE PROGRAM

Python:

```
Print("Hello World")
```

Output: Hello World

C:

```
#include <stdio.h>
{
    printf("Hello World");
}
```

Output: Hello World

I. Save the Python file

Save the python file with the extension ".py"

Example: *Filename.py*

II. Run

Python Filename.py

THANK YOU